

PATIALA QUICK-COMMERCE ETA VOLATILITY & SERVICEABILITY AUDIT

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1. Executive Summary & Core Operational Discoveries

This case study presents an end-to-end operational investigation into quick-commerce delivery dynamics across Patiala, Punjab. By gathering primary telemetry across 78 live operational checkpoints from 10:02 AM to 10:30 PM across three distinct micro-markets (Thapar University, Model Town, and New Bus Stand), this audit diagnoses structural serviceability failures and ETA inflation during evening peak hours.

Key Analytical Findings:

- Catchment Uptime Divergence:** *New Bus Stand* maintained 100% serviceability (0% stockout) and a tight 8–13 minute delivery window continuously across all 26 monitored time slots.
- Evening Service Collapse:** *Thapar University* and *Model Town* experienced severe ETA spikes (up to 30 mins) and surge pricing during the dinner rush (8:00–8:34 PM), followed by a **100% serviceability collapse (0% item availability)** from 8:30 PM to 10:30 PM.
- Operational Hypothesis:** The observed ETA and serviceability divergence may indicate differences in local fulfillment or delivery capacity; however, internal order, inventory and rider data would be required to determine the underlying cause.

2. Primary Telemetry Methodology & Data Engineering Pipeline

To evaluate real-world quick-commerce performance without relying on secondary Kaggle benchmarks, a primary telemetry protocol was executed directly on the Blinkit customer application interface. A standardized 11-item grocery basket valued at Rs. 533 (comprising household staples) was monitored every 30 minutes across three geographically separated Patiala localities.

Telemetry Parameter	Logged Data Specification	Operational Relevance
Temporal Sampling	26 Time Windows (10:02 AM – 10:30 PM)	Captures full daily operating cycle & peak shifts
Spatial Micro-Markets	Thapar Univ. Model Town New Bus Stand	Compares student, residential, & commercial hubs
Standard Basket	11 Fixed Items	Controls for basket size bias on pick/pack times
Recorded Metrics	ETA (mins), Available Items, Surge, Weather	Quantifies fulfillment speed & capacity bottlenecks

3. Descriptive Telemetry Observations & Locality Analytics

Locality Name	Total Logs	Active ETA Range	Average ETA	Surge Events	Serviceability %
New Bus Stand Patiala	26 Slots	8 – 13 mins	10.23 mins	0 Events	100.0% Uptime
Model Town	26 Slots	11 – 23 mins	15.90 mins	3 Events	80.8% (5 Outages)
Thapar University	26 Slots	11 – 30 mins	18.87 mins	4 Events	88.5% (3 Outages)

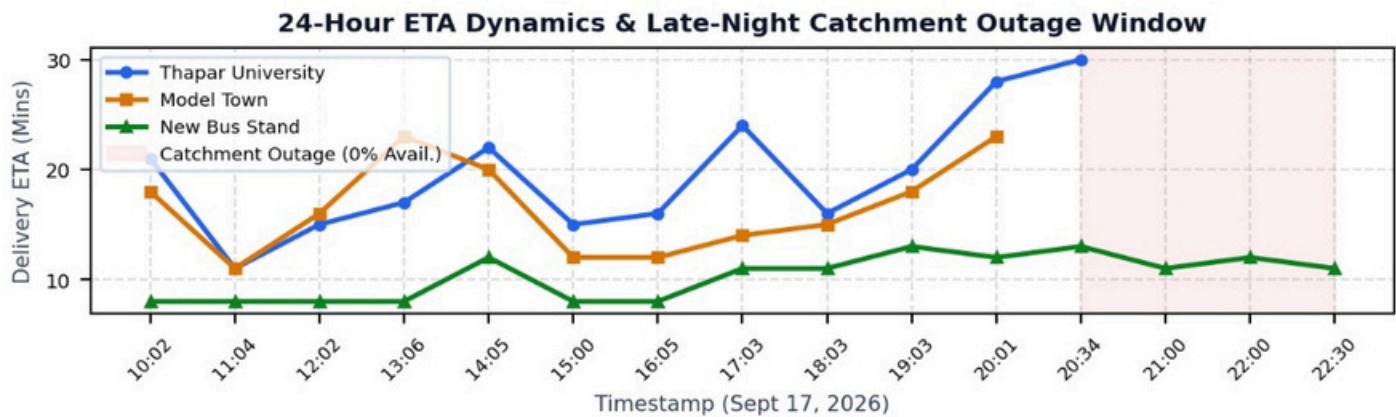


Figure 1: 12-Hour Telemetry Tracking across 3 Patiala Localities highlighting the post-20:30 PM Catchment Collapse.

Temporal Surge Window Analysis: Surge pricing was triggered strictly across two distinct high-demand windows: 1) *Evening Tea/Snack Rush* (5:03 PM – 5:40 PM) where Thapar University and Model Town ETAs jumped to 24m and 20m respectively; 2) *Dinner Prep Peak* (8:01 PM – 8:34 PM) where Thapar University peaked at 30 minutes and Model Town hit 23 minutes. In contrast, New Bus Stand maintained a flat 11–13 min ETA without incurring surge fees.

4. Logical Root Cause Analysis (RCA): The 5-Whys Framework

Demand Trigger: Why does evening demand typically intensify?

The evening period combines post-work/student activity, household replenishment and meal-related shopping, creating a concentrated demand window rather than evenly distributed demand throughout the day.

Locality Concentration: Why are some catchments affected more than others?

Demand intensity can differ by population density, customer demographics, land use and consumption patterns. Thapar University, Model Town and New Bus Stand represent materially different customer environments, which can produce different order-demand profiles.

Fulfillment Load: Where can the demand spike create friction?

A concentrated order surge increases simultaneous picking, packing and dispatch workload at the stores serving those catchments. Blinkit states that its technology enables partners to pick and pack orders rapidly, making store-level throughput an important component of the delivery promise.

Last-Mile Capacity: What happens when delivery demand rises faster than available capacity?

Higher concurrent orders can increase pressure on the available delivery-partner pool and travel capacity. Blinkit describes delivery partners as independent contractors who choose working schedules, making time-of-day availability another operational variable worth investigating.

Root-Cause Hypothesis: Why can the same demand spike produce different locality outcomes?

The observed divergence may reflect differences in local demand intensity, store throughput, delivery-partner availability, service-area design and travel-time conditions. Determining the actual bottleneck would require internal order, rider, store-capacity and routing data.

5. Study Limitations & Telemetry Scope Boundaries

- **Single-Day Operational Snapshot:** Telemetry was captured on Sept 17, 2026. While capturing 26 time slots provides high temporal detail, longitudinal multi-day sampling is required to measure day-of-week seasonality.
- **Black-Box Dark Store Internal Logs:** As a customer-side primary study, metrics reflect outward SLA promises and availability rather than internal WMS picking speeds, batching ratios, or inventory bin stock levels.
- **Rider Trajectory Estimation:** Delivery ETAs reflect total promised fulfillment times. Distinguishing order prep time from transit travel time requires internal telemetry access.